



NORTH SOUTH UNIVERSITY
Department of Electrical and Computer Engineering
B.Sc. in Computer Science and Engineering Program
Final Assessment (Assignment), Spring 2020 Semester

Course: CSE 225 Data Structure and Algorithms
Instructor: Mohammad Rezwanul Huq (MRH1), PhD, Assistant Professor (Part-time)
Full Marks: 50 (25 will be counted for final grading)
Submission Deadline: 04 June 2020, 11:59PM

Note: There are 5 (FIVE) questions, answer ALL of them. Mark of each question are mentioned at the right margin.

Special Instructions:

1. Write your answer in white pages.
 2. Each page must have your student ID, name, signature, page number and Set number in the top.
 3. Take photo of each page and make one PDF file out of them. Name the PDF file after your student id such as 2016_1_60_001.pdf. Attach the given cover page as top sheet in your PDF file with.
 4. If you are unable to make a PDF, then rename each photo like your_student_id_page1.jpg, your_student_id_page2.jpg and so on.
 5. Upload the file(s) before the deadline. Do not zip the file(s).
 6. Please Click 'Mark as done'.
 7. Submission deadline is 04 June 2020, 11:59PM.
 8. Late submission by each 10 minutes will deduct 2 marks. After 150 minutes, submissions will no longer be considered, and you will get no points for this assessment. Even if you are late by 1 minute, I will deduct your marks.
 9. If plagiarism is detected in students answers, i.e. same wordings (direct copy-paste), 50% marks of that question will be deducted automatically. I will strictly follow that.
- Good Luck.

1. a) The postorder traversal of a Binary Search Tree (BST) is given below. [10]

40 60 10 15 75 120 130 150 110 70

Draw the BST and briefly explain the process you followed to build the tree.

b) Write a function `f1` that takes the root of a binary tree as a parameter and returns the sum of the nodes which are in the right subtree of the root. You must not consider the value of the root in the final sum. Consider the following class definitions while writing your code.

```
class Node {
public:
    int key;
    Node* left;
    Node* right;
};
```

2. a) Consider the following AVL Tree. [8]

```
      60
     /  \
    30   75
     \
     40
```

Insert the following values in the given AVL Tree, one after another, and **show** the resulting tree after each insertion. You must justify your answer by explaining the rotation operation you performed during insertion.

50 45 80 120 110 17

b) Delete the root of the resulting tree in Question 2.a. **Show** the resulting AVL Tree. Justify your answer by showing every step during deletion.

3. **a) Draw** the binary max-heap after inserting the following values, one after another. [12]

15, 2, 32, 40, 25, 36, 80, 100

You must show each step of building the heap and eventually the final tree. *Please, put your final tree inside a box so that it can be easily understood among other intermediate trees.*

b) A 4-ary min heap is like a binary min heap, but instead of 2 children, nodes have 4 children. A 4-ary heap can be represented by an array as shown in Figure 1 (up to level 2).

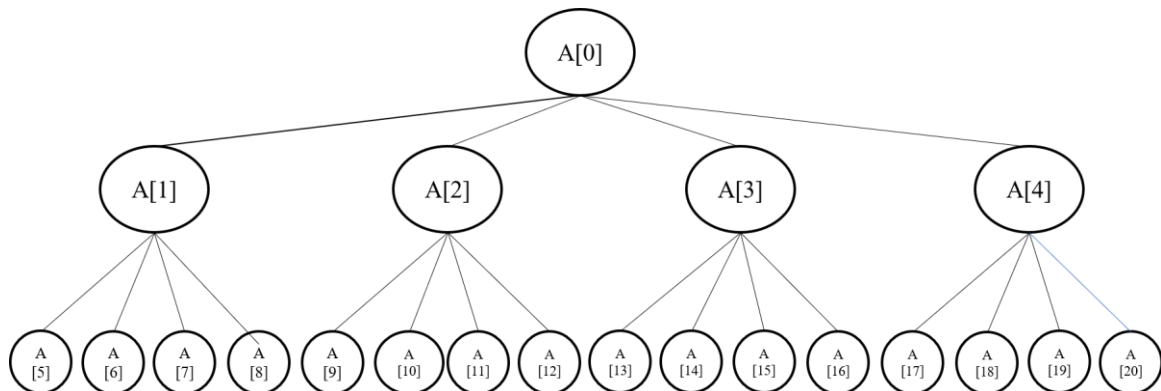


Figure 1: 4-ary heap representation using an array

Write down `MIN_HEAPIFY(A, i)` function for a 4-ary min-heap that restores the heap property for i^{th} node.

4. **a) Write** a program that asks user to enter number of vertices in a directed graph and then the adjacency matrix representing the directed graph. The program, then, must display the node with the highest outdegree. Assume that nodes are named as 0, 1, 2 and so on. [12]

You will find a sample run of the program below.

```

Enter number of vertices: 4
Enter adjacency matrix:
0 1 1 1
0 0 0 0
0 0 0 0
0 1 1 0

Node with the highest outdegree: 0
  
```

b) Consider the following undirected graph as shown in Figure 2. Find the articulation points in the given graph. Justify your answer.

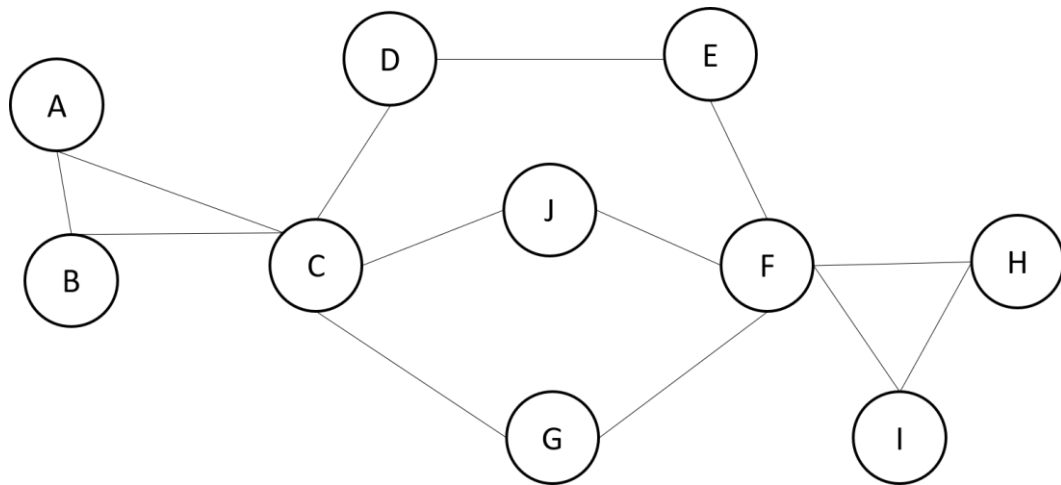


Figure 2: An undirected graph for Question 4.b

5. Consider the following weighted undirected graph as shown in Figure 3. **Find** the shortest path distance from node A to all other nodes. **Show** calculation of each iteration while finding the shortest path. [8]

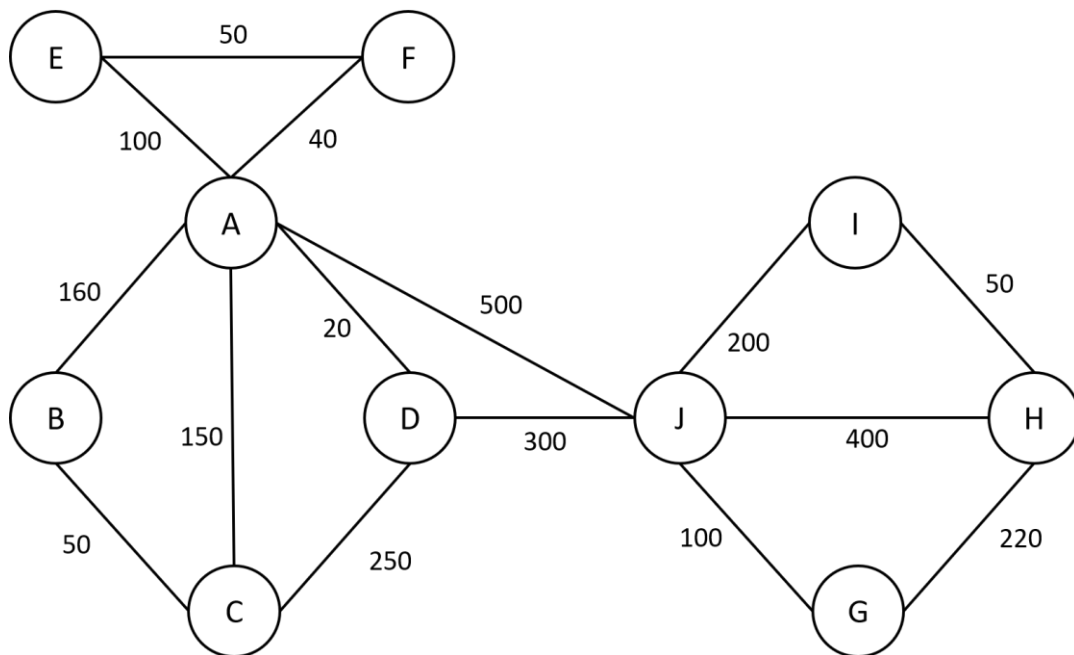


Figure 3: A weighted undirected Graph for Question 5